Body Control System

High-Level Design (HLD)

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This document describes the software architecture of the body control domain: exterior lighting, climate control, seat comfort and access management components, ports, interfaces, signals and functional flows.

1 Revision history

Revision	Date	Author	Description
B	2026-05-14	A. Kumar	Initial architecture baseline

2 Scope and context

2.1 Scope

The body control domain covers exterior lighting commands, cabin climate regulation, seat comfort actuators and keyless access state management. Out of scope: actuator driver electronics and production wiring harness details.

2.2 System context

The domain runs on BodyEcu and communicates with the vehicle motion domain for lamp commands and speed signals through BodyGateway over the body CAN bus at 125 kbit/s.

3 Software architecture

3.1 Component breakdown

Component	Type	Description
ExteriorLightingController	ApplicationSwComponentType	Headlamp and indicator logic
ClimateControlUnit	ApplicationSwComponentType	Cabin temperature regulation
SeatComfortManager	ApplicationSwComponentType	Seat adjustment and comfort featur
AccessManagementUnit	ApplicationSwComponentType	Keyless access and authorization
WiperController	ApplicationSwComponentType	Wiper and washer automation
BodySignalService	ServiceSwComponentType	Body signal gateway service
BodyGateway	EcuAbstractionSwComponentType	Cross-domain gateway exchange
BodyControlComposition	CompositionSwComponentType	Composition of the body components

3.2 Ports and interfaces

Port	Direction	Interface	Component
LampCommandIn	required	LampCommandI	ExteriorLightingController
IndicatorStateOut	provided	IndicatorStateI	ExteriorLightingController
CabinTemperatureOut	provided	CabinTemperatureI	ClimateControlUnit
SeatPositionOut	provided	SeatPositionI	SeatComfortManager
AccessStateOut	provided	AccessStateI	AccessManagementUnit
BodyGatewayIn	required	BodySignalSetI	BodySignalService
GatewayToMotion	provided	LampCommandI	BodyGateway
WiperStateOut	provided	WiperStateI	WiperController

3.3 Signal inventory

Signal	Interface	Sender	Receiver	Bits
LampCommand	LampCommandI	BodyGateway	ExteriorLightingContr	8
IndicatorBlink	IndicatorStateI	ExteriorLightingControl	WiperController	1
CabinTemperature	CabinTemperatureI	ClimateControlUnit	BodySignalService	8
SeatMemoryIndex	SeatPositionI	SeatComfortManager	AccessManagementUnit	4
AccessGranted	AccessStateI	AccessManagementUnit	ClimateControlUnit	1
RainSensorValue	RainSensorI	WiperController	ExteriorLightingContr	8
AmbientLightLevel	AmbientLightI	BodyGateway	ExteriorLightingContr	8

4 Functional flows

4.1 Exterior lighting flow

BodyGateway forwards LampCommand from the vehicle motion domain to ExteriorLightingController. ExteriorLightingController consumes AmbientLightLevel and RainSensorValue to decide headlamp state and publishes IndicatorStateOut.

4.2 Access and comfort flow

AccessManagementUnit authorizes the driver key and publishes AccessStateOut. ClimateControlUnit consumes AccessGranted to begin pre-conditioning. SeatComfortManager applies the stored SeatMemoryIndex for the authorized driver.

4.3 Wiper automation flow

WiperController consumes RainSensorValue and publishes WiperStateOut. WiperController requests reduced wiper activity during vehicle standstill while StartStopStateI is active.

5 Timing and runnables

TimingEvent LightingUpdateEvent triggers UpdateLightingRunnable every 50 ms. TimingEvent ClimateControlEvent triggers RegulateClimateRunnable every 200 ms. TimingEvent AccessPollEvent triggers PollAccessStateRunnable every 100 ms. All runnables are scheduled by BodyTask on BodyEcu.

6 Basic software integration

Signal routing uses Com and PduR. Incoming frames are handled by CanIf on BodyEcu. NvM stores the seat memory and lighting preference blocks. Rte provides the port API used by all body application components.

Mode management is performed by BswM and EcuM. Watchdog supervision for BodyTask is performed by WdgM. The wiper motor interface uses IoHwAb for pulse width modulation control.

The diagnostic path routes UDS requests through Dcm and Dem. BodySignalService consumes BodySignalSetI and forwards routine results to the tester.

7 Diagnostics and configuration

DTCs for lamp circuit failure, climate sensor plausibility and access antenna fault are defined in the diagnostics specification. The HLD only defines the reporting path: component to Dcm to CanIf.

Configuration parameters are grouped into the domain configuration container and written through NvM during shutdown.

8 Known limitations

SeatComfortManager supports two memory positions per seat in this revision; more positions require a NvM layout change.

9 Open questions

Open: calibration ownership for cross-domain interfaces and the timeline for production software freeze are tracked outside this document.